

Programme	:	Bachelor of Computer Science (Data Engineering)
Subject Code	:	SECJ 2154
Subject Name	:	OBJECT-ORIENTED PROGRAMMING
Session-Sem	:	2024/2025-2
MINI PROJECT TITLE	:	MyNoodle food ordering system (PROPOSAL)

Prepared by 1. WOO CHENG SHUAN (A23CS0283)
: 2. ELIJAH SHE YU SHENG (A23CS0073)
3. MUHAMMAD MUKHRITZ AL (A23CS0)
IMAN BIN MOHD

Section : 02

Lecturer : Dr. Zuraini Ali Shah

Date : 07-06-2025

TABLE OF CONTENT

No.	Section Title	Page
1.0	Introduction	3
2.0	Problem statement	3
3.0	Class Diagram of the project	4
	3.1 The Class Diagram Description	5
4.0	Technique//Software//Tools being used	6

1.0 Introduction

The MyNoodle Food Ordering System is designed to make it easier and faster to handle the whole process of ordering, preparing and delivering noodle dishes. It uses object-oriented programming to help manage things like creating and customizing the menu, handling customer profiles and their preferences, placing and tracking order in real time, processing payments safely, and organizing deliveries.

This system built to grow with the business and stay reliable even when handling many orders. It helps improve how the kitchen runs, keep track of stock and sale correctly and gives a smooth and user-friendly experience for both customer and kitchen staff.

2.0 Problem statement

Inefficient practices like long wait times, lost orders, inaccurate inventory tracking and inconsistent customer experiences can result from traditional noodle shops' frequent reliance on manual procedures for order taking, menu management and delivery tracking. Scaling operations and preserving service quality during peak hours are made more difficult by these constraints.

The MyNoodle Food Ordering System is proposed to address these challenges by introducing an automated, end-to-end digital solution to address these issues. Customizable menu administration, order placement, payment processing and tracking in real-time are some of the main features. By transitioning from manual to automated workflows, the system seeks to reduce errors, optimize kitchen and delivery operations, improve customer satisfaction and provide accurate inventory and sales insights.

3.0 Class Diagram of the project

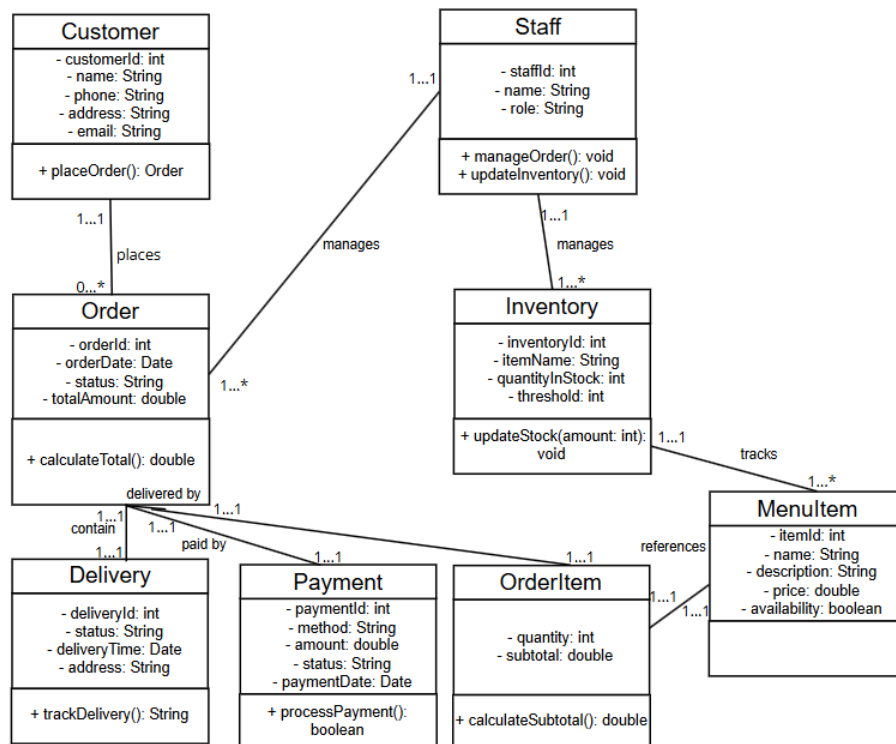


Table 3.0.1 MyNoodle Food Ordering System Class Diagram

The MyNoodle Food Ordering System is designed using object-oriented principles with interconnected classes to manage the full ordering workflow. The Customer class allows users to place orders, which are handled by the Order class containing multiple OrderItem instances. Each OrderItem is linked to a MenuItem representing dishes on the menu. The Order is associated with both a Payment for transaction details and a Delivery to track the delivery status. Inventory control is managed through the Inventory class, which monitors stock levels of menu items. The Staff class oversees both order and inventory management, ensuring smooth kitchen and delivery operations. All classes are interlinked to support real-time ordering, payment, and delivery in a seamless and automated manner.

3.1 The Class Diagram Description

Class Name	Description	Key Attributes	Relationships
Customer	Represents a user who places food orders.	customerId, name, phone, address, email	- Places one or more Order
MenuItem	Represents an individual item on the menu.	itemId, name, description, price, availability	- Referenced by OrderItem- Tracked by Inventory
Order	Represents a customer's complete order.	orderId, orderDate, status, totalAmount	- Placed by Customer- Contains OrderItem- Linked to one Payment and one Delivery- Managed by Staff
OrderItem	Represents each individual item within an order.	quantity, subtotal	- References one MenuItem- Part of one Order
Payment	Records payment information for an order.	paymentId, method, amount, status, paymentDate	- Linked to one Order
Delivery	Tracks the delivery status of an order.	deliveryId, status, deliveryTime, address	- Linked to one Order
Inventory	Tracks menu ingredients or item stock levels.	inventoryId, itemName, quantityInStock, threshold	- Tracks multiple MenuItem- Managed by Staff
Staff	Represents an admin or kitchen/delivery worker.	staffId, name, role	- Manages multiple Order and Inventory

4.0 Technique//Software//Tools being used

State or tick the main concept of Java Programming to be used in this project

☐ ENUM	☒	ASSOCIATION*
☒ ARRAY LIST /.VECTOR	☐	COMPOSITION
☐ ARRAY	☐	AGGREGATION
☒ ARRAY OF OBJECT	☒	INHERITANCE*
☒ FILE*	☐	STATIC METHOD
☒ POLIMORPHISM*	☒	ABSTRACT CLASS*
☒ INTERFACE CLASS*	☒	EXCEPTION HANDLING*
☐ GUI APPLLCATION	☐	Others: _____

- ***compulsory – also comments, style etc**
- State tools, apps or software to support this mini project.